

A Real-time approach to improve drilling decision-making process using virtual reality visualization

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ABSTRACT

Virtual reality (VR) is one of the key Industry 4.0 trends and is being largely used for training and simulations. A VR environment can reduce training and drilling analysis costs, and help operators and coordinators to monitor the trajectory and other operational variables during the drilling process. This paper presents Divisor, a virtual reality tool for monitoring variables and analyzing historical and real-time data while drilling a new oil well.

Index Terms: Human-centered computing—Virtual reality—Visualization techniques—; Drilling—monitoring analysis

1 INTRODUCTION

The drilling of an oil well consists of numerous phases, such as drilling, casing installation, and cementing, with the reduction of the well's diameter at each stage until reaching the reservoir [5]. Moreover, in the particular case of offshore drilling, the mechanical process is performed below the seabed. The process consists of the drill rig, drill pipes, bottom hole assembly (BHA) to drill in this context (5 in Fig. 1). During this process, operational variables and the well's trajectory are monitored in real-time.

Monitoring a well's trajectory is an activity that simulates countless scenarios, such as depth of the phases and types of casing [4] using specific software. Consequently, the accurate information can help the drilling team choose the best equipment, accessories, and operating conditions for an oil well. This monitoring reduces numerous problems and risks during drilling activities and maneuvers [3]. However, most of this software does not have an VR or 3-dimensional view of the data [7], [4].

Another point of attention is the cost. According to Kaiser et al. [6], well drilling is one of the most costly activities in oil and gas exploration. Additionally, in offshore fields, rental equipment and personnel are expensive, and drilling times are longer. On the other hand, we are experiencing the consolidation of new training platforms using virtual environments. Virtual reality allows the visualization of realistic 3D graphics in training applications for industry, science, or military areas. For this reason, VR can be a way to reduce costs in drilling activities. This project aims to read, organize, and integrate data generated by different simulation software used in the planning phase with real data obtained during drilling operations for real-time visualization with VR's aid.

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1.1 Related Works

Brasil et al. [2] presents a serious game that simulates a drilling rig from the educational side. In this game, users can exercise drilling activities and learn what they will need to do in dangerous situations. Mostafa et al. [9] propose JackVR, an immersive simulation and semi-realistic training environment for the process of landing oil rigs. Leta et al. [8] present a framework for visualizing and reconstructing the geometry of drilling wells. The developed tool is capable of visualizing wells with thousands of meters. The system also has movement, zoom, rotation, and translation functionality.

Our work focuses on data visualization related to geometry, trajectory and operational variables monitoring to help drilling engineering teams' decision-making. Using historical data it could also be used for creating training scenarios and after action simulations. We use the 3D model reconstruction presented by Leta et al. to create our real-time 3D visualization of the well's trajectory.

2 MATERIALS AND METHODS

We developed a 3D visualization system to help the drilling monitoring and analysis process. The system has two usability forms: non-VR and VR (Figure 1). A unified database in the cloud provides data from multiple sources to our application. Data fetched from this database is used to generate a 3D visualization of the well and the BHA, to plot historical and real-time variable data that can be monitored and associated with custom alarms, and to visualize diagnostics information generated by artificial intelligence (AI).

The application was developed in Unity 3D using SteamVR, which makes it compatible with head-mounted displays (HMDs), such as HTC Vive and Oculus Rift, when running in VR mode. While these systems provide highly immersive experiences, many users have reported discomfort symptoms, such as nausea, sickness, and headaches, among others. We implement some strategies to overcome cybersickness [10] in our application, such as teleporting and mesh optimization to keep the frame rates stable, using the method described by Leta et al. [8] to generate the geometry.

3 DIVISOR APPLICATION

The application operates in two ways, the non-VR mode, where the user views the content on a computer screen, and the VR mode, where the user uses a virtual reality headset to view the content. The VR mode allows users access drilling data in an immersive environment that can be customized as needed to better display all the drilling variables that need to be monitored or analyzed in real-time or in a review session. It can also be used as a training environment using historical data that should be monitored and analyzed by the user.

First, the user can select which drilling project to load from the central application controller (2 in Fig. 1). After the wellbore was loaded entirely (1 in Fig. 1), the application allows visualizing the drilling data in different ways, such as dashboards, color mapping or navigating through historical data.

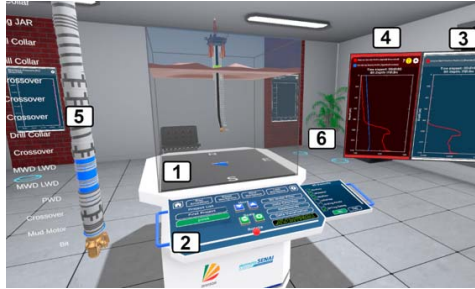


Figure 1: The Divisor VR environment. (1) The wellbore 3D visualization. (2) The main application controller. (3) Dashboard monitoring. (4) Dashboard alarming. (5) BHA equipment's visualization. (6) Teleporting point.

3.1 Dashboards

The Dashboard tool enables the operator to create panels and monitor different operational variables during the drilling process in real-time or to visualize historical data. These dashboard elements can be customized in terms of graph color and multi variable display (Fig. 2). Additionally, it is possible to set alarms that are triggered when a value is above or below a given limit (4 in Fig. 1).



Figure 2: Dashboard tool monitoring fluid density drilling data.

3.2 Color Mapping

The data can also be visualized as a color map, ranging from min to max values using two colors set by the user. This enables a color map visualization, which means a color representation values set tied to the monitored variable data (Fig. 3). It's possible to set color mapping for a curve as local (using min/max values at the current time), global (using min/max values between all available data for that curve) or custom (using min/max values as set by the user).

3.3 Historical data and interactions

The application allows the user to access historical data or to visualize real-time updates. Concerning the movements, users can move freely in the virtual environment. Moreover, to help move between locations, some teleporting points are positioned in the VR environment (6 in Fig. 1). The teleporting points allow the user to get closer to a fixed dashboard and visualize data from up close. Additionally, the user can manipulate the 3D objects and dashboard elements around the virtual environment using the grab/pick functions.

4 CONCLUSION

We developed a virtual reality visualization tool that enables the user to analyze drilling data in VR (with the option of using a non-

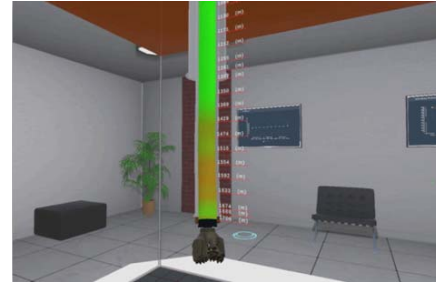


Figure 3: Color mapping tool - visualization of pressure along the bore whole depth.

VR alternative). It allows the user to navigate the virtual environment and 3D models (such as platform, drill, and others), visualize data in multiple ways, perform assessments, training, and simulations. Moreover, the VR application enables the user presence sensation that does not exist in the non-VR software. Additionally, this tool offers a low-cost real-time monitoring tool to improve drilling decision-making process using VR visualizations. Future work includes extending the tool with more visualizations in VR, and non-VR modes, including data generated by AI and drilling projects comparison (real and simulated) integrating it to a company-wide Digital Twin platform [1].

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